Dr. Dhruv Kumar

Assistant Professor, BITS Pilani

LLMs • Agentic AI • Recursive Self-Improvement • Autonomous AI for Science

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Personal Details

Field	Details
Name	Dr. Dhruv Kumar
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Academic Qualifications

#	Degree	Year	Subject	University / Institution	CGPA
1	B.E. (Hons.)	2014	Computer Science	Birla Institute of Technology and Science, Pilani	9.92/10
2	Ph.D.	2022	Computer Science & Engineering	University of Minnesota, Twin Cities, USA	4.0/4.0

Work Experience

#	Position	Institution	From	To
1	Assistant Professor	BITS Pilani	May 2024	Present
2	Adjunct Faculty	IIIT Delhi	May 2024	Present
3	Assistant Professor	IIIT Delhi	Aug 2022	May 2024
4	Post-Doctoral Researcher	Microsoft Research, Bengaluru	Mar 2022	Jul 2022
5	Ph.D. Intern	Google Cloud, California, USA	May 2019	Aug 2019
6	Software Engineer / Quant Analyst	Goldman Sachs, Bengaluru	Nov 2014	Apr 2016
7	Summer Intern	CSIR-CEERI, Pilani	May 2012	Jul 2012

Professional Recognition / Awards / Prizes / Fellowships

#	Award / Recognition	Awarding Body	Year
1	Teaching Excellence Award (AY 2023–24)	IIIT Delhi	Sep 2024
2	Teaching Excellence Award (AY 2022–23)	IIIT Delhi	Sep 2023
3	3M Science and Technology Fellowship	University of Minnesota, Twin Cities	2017–2021
4	Travel Grant	USENIX ATC	2019
5	Travel Grant	ACM SIGMETRICS	2019
6	Travel Grant	USENIX OSDI	2018
7	Merit Scholarship (Top 10 among 800 students)	BITS Pilani	2010–2014
8	Outstanding Undergraduate Thesis	BITS Pilani	2014

Publications (in Year-wise Descending Order)

*Author's name in **bold**. [Best Paper Award] denotes prize-winning papers.*

Journal Articles & Transactions

J1. Alakh Sharma, Gaurish Trivedi, Kartikey Singh Bhandari, Yash Sinha, **Dhruv Kumar**, Pratik Narang, and Jagat Sesh Challa. 2026. Generative Evolutionary Meta-Solver (GEMS): Scalable Surrogate-Free Multi-Agent Reinforcement Learning. **Transactions on Machine Learning Research (TMLR) 2026**.

J2. Raj Gupta, Harshita Goyal, **Dhruv Kumar**, Apurv Mehra, Sanchit Sharma, Kashish Mittal, and Jagat Sesh Challa. 2026. Sakshm AI: Advancing AI-Assisted Coding Education for Engineering Students in India Through Socratic Tutoring and Comprehensive Feedback. **ACM Journal on Computing and Sustainable Societies (JCSS) 2026**.

J3. Harshita Goyal, Garima Garg, Prisha Mordia, Veena Ramachandran, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. The Impact of Large Language Models on K-12 Education in Rural India: A Thematic Analysis of Student Volunteers' Perspectives. **Scientific Reports (2025)**.

J4. Balamurugan Palanisamy, Vikas Hassija, Arpita Chatterjee, Arpita Mandal, Debanshi Chakraborty, Amit Pandey, G. Sai Sesha Chalapathi, and **Dhruv Kumar**. 2025. Transformers for Vision: A Survey on Innovative Methods for Computer Vision. **IEEE Access 13 (2025)**, 95496–95523.

J5. Harshal Akolekar, Piyush Jhamnani, Vikash Kumar, Vinay Tailor, Aditya Pote, Ankit Meena, Kamal Kumar, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. The Role of Generative AI Tools in Shaping Mechanical Engineering Education from an Undergraduate Perspective. **Scientific Reports 15 (2025)**, Article 9214.

J6. Ashutosh Sharma, **Dhruv Kumar**, Srinivas Rallapalli, et al. 2023. Wetland Functional Assessment and Uncertainty Analysis Using Fuzzy Alpha-Cut–Based Modified Hydrogeomorphic Approach. **Environmental Science & Pollution Research 30 (2023)**, 72900–72915.

J7. Ananya Jain, Srinivas Rallapalli, and **Dhruv Kumar**. 2022. Cloud-Based Neuro-Fuzzy Hydro-Climatic Model for Water Quality Assessment under Uncertainty and Sensitivity. **Environmental Science & Pollution Research 29 (2022)**, 65259–65275.

J8. Poonam Goyal, Jagat Sesh Challa, **Dhruv Kumar**, Anuvind Bhat, Sundar Balasubramaniam, and Navneet Goyal. 2020. Grid-R-Tree: A Data Structure for Efficient Neighborhood and Nearest Neighbor Queries in Data Mining. **Int. J. Data Sci. Anal. 10 (2020)**, 25–47.

J9. **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics. **Proc. ACM MACS 3, 2 (2019)**, 29:1–29:27.

J10. **Dhruv Kumar**, Poonam Goyal, and Navneet Goyal. 2018. An Efficient Method for Batch Updates in OPTICS Cluster Ordering. **Int. J. Data Anal. Tech. Strateg. 10 (2018)**, 57–80.

Conference Papers

C1. Aaryaman Kalani, Murari Mandal, **Dhruv Kumar**, Mohan Kankanhalli, and Yash Sinha. 2026. The BatchNorm Illusion: Diagnosing Normalization Artifacts in Machine Unlearning Evaluation. In **NeurIPS 2026**.

C2. Stuti Sinha, Himanshu Kumar, Aryan Raju Mandapati, Rakshit Sakhuja, and **Dhruv Kumar**. 2026. The Language of Bargaining: Linguistic Effects in LLM Negotiations. In **AACL-IJCNLP 2026 (Main Conference)**, also accepted at **ICLR 2026 Workshop (LLM Reasoning)**, **ACL 2026 Workshop (C3NLP)**, **ICML 2026 Workshop (Culture x AI)**, and **ICML 2026 Workshop (Pluralistic Alignment)**.

C3. Darshil Chauhan, Adityasinh Solanki, Vansh Patel, Kanav Kapoor, Ritvik Jain, Aditya Bansal, **Dhruv Kumar**, and Pratik Narang. 2026. Navigating the Reality Gap: Privacy-Preserving On-Device Continual Adaptation of ASR for Clinical Telephony. In **AACL-IJCNLP 2026 (Main Conference)**.

C4. Dhruv Gupta, Vasudev Majhi, Jagat Sesh Challa, Murari Mandal, and **Dhruv Kumar**. 2026. Anchored Turns: A Length-Driven Gradient in Chain-of-Thought Causal Necessity Across Multi-Turn Conversations. In **AACL-IJCNLP 2026 (Main Conference)**.

C5. Venkat Aniruddh Thamatam, Vasudev Majhi, Aaryan Nagda, Jagat Sesh Challa, Murari Mandal, Yash Sinha, and **Dhruv Kumar**. 2026. Method Ranking Inverts Across Targets: A Matched-Grid Evaluation of Jailbreak Attacks on Open-Weight Vision-Language Models. In **AACL-IJCNLP 2026 (Main Conference)**.

C6. Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Jahnvi Singh, Vinay Chamola, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2026. When Reject Turns into Accept: Quantifying the Vulnerability of LLM-Based Scientific Reviewers to Indirect Prompt Injection. In **Findings of AACL-IJCNLP 2026**.

C7. Shiven Sharma, Gautam Choudhary, Ashish Sam, and **Dhruv Kumar**. 2026. Benchmarking Frontier Agentic Deception in LLMs vs Humans. In **Findings of AACL-IJCNLP 2026**.

C8. Debdeep Sanyal, Aakash Sen Sharma, **Dhruv Kumar**, Saurabh Deshpande, and Murari Mandal. 2026. Format Collapse in Policy Optimization for LLM Reasoning. In **EMNLP 2026**.

C9. Soham Roy, Sarthakbrata Halder, Arya Bharaty, Vaibhav Bhaskar, Yash Sinha, **Dhruv Kumar**, Srikant Panda, and Murari Mandal. 2026. "I Strongly Suspect This Website Is a Scam": Benchmarking PII Leakage and Detection without Defense in Autonomous Web Agents. In **EMNLP 2026**.

C10. Debdeep Sanyal, Umakanta Maharana, Hong Ming Tan, Yash Sinha, Shirish Karande, Mohan Kankanhalli, **Dhruv Kumar**, and Murari Mandal. 2026. OrgAccess: A Benchmark for Role-Based Access Control in Organization Scale LLMs. In **EMNLP 2026**.

C11. Sasank Tumpati, Devansh Agarwal, Ayush Kedia, Arjun Neekhra, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, and **Dhruv Kumar**. 2026. FinBalance: A Multi-Document Accounting Reconciliation Benchmark. In **Findings of EMNLP 2026**.

C12. Aniket Upadhyay, Aakash Trivedi, Sparsh Bajoria, Shreyanshu Ranjan, **Dhruv Kumar**, and Pratik Narang. 2026. PRAXIS: A 68-Industry Benchmark for Pragmatic-Form Bias in Suggestion Mining. In **Findings of EMNLP 2026**.

C13. Aniket Upadhyay, Aakash Trivedi, **Dhruv Kumar**, and Pratik Narang. 2026. Certified Cost-Optimal Triage for Suggestion Detection in Reviews. In **EMNLP 2026 Industry Track**.

C14. Tulika Tewari, Nalin A. G. Arachchilage, Jagat Sesh Challa, and **Dhruv Kumar**. 2026. From Trust to Compromise: Outcome-Verified LLM Phishing Simulation and Real-Time Defense. In **Association for Computational Linguistics (ACL 2026)**.

C15. Samyak Savi, Chavi Gupta, Shreyas Gantayet, Tanay Sodha, and **Dhruv Kumar**. 2026. Evaluating and Mitigating Misgendering in English-to-Hindi Machine Translation. In **COLM 2026**, also accepted at **ICML 2026 Workshop (Culture x AI)**.

C16. **Dhruv Kumar**. 2026. Oracle-Agnostic Text Quality Optimization via Prompt-Only Agentic Loops. In **CAISc 2026**.

C17. Vikranth Udandarao, Noel Abraham Tiju, Muthuraj Vairamuthu, Harsh Mistry, and **Dhruv Kumar**. 2026. Roamify: Designing and Evaluating an LLM-Based Google Chrome Extension for Personalised Itinerary Planning. In **The Web Conference (WWW) 2026**.

C18. Pankaj Iiitd, **Dhruv Kumar**, Vinayak Abrol, and Vikram Goyal. 2026. BLEND: Balanced and Leaf-Enhanced Dual Fine-Tuning for Taxonomy Completion. In **The Web Conference (WWW) 2026**.

C19. Aakash Trivedi, Aniket Upadhyay, Pratik Narang, **Dhruv Kumar**, and Praveen Kumar. 2026. A Hybrid Supervised-LLM Pipeline for Actionable Suggestion Mining in Unstructured Customer Reviews. In **EACL 2026 Industry Track**.

C20. Debdeep Sanyal, Agniva Maiti, Umakanta Maharana, **Dhruv Kumar**, Ankur Mali, C. Lee Giles, and Murari Mandal. 2025. Investigating Pedagogical Teacher and Student LLM Agents: Genetic Adaptation Meets Retrieval-Augmented Generation Across Learning Styles. In **EMNLP 2025**, 13359–13400.

C21. Varun Bharti, Shashwat Jha, **Dhruv Kumar**, and Pankaj Jalote. 2025. Combining Reasoning Optimised LLMs and SMT Solvers for Automated Loop Invariant Synthesis. In **AIWare 2025**. **[Best Paper Award]**

C22. Aditya Pathak, Rachit Gandhi, Vaibhav Uttam, Arnav Ramamoorthy, Pratyush Ghosh, Aaryan Raj Jindal, Shreyash Verma, Aditya Mittal, Aashna Ased, Chirag Khatri, Yashwanth Nakka, Devansh, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. Rubric Is All You Need: Improving LLM-Based Code Evaluation With Question-Specific Rubrics. In **ICER 2025**, 181–195.

C23. Adithya Chittem, Aishna Shrivastava, Sai Tarun Pendela, Jagat Sesh Challa, and **Dhruv Kumar**. 2025. SAC: A Framework for Measuring and Inducing Personality Traits in LLMs with Dynamic Intensity Control. In **ICAART 2025**.

C24. Madhav Krishan Garg, Tejash Prasad, Tanmay Singhal, Chhavi Kirtani, Murari Mandal, and **Dhruv Kumar**. 2025. ReviewEval: An Evaluation Framework for AI-Generated Reviews. In **Findings of EMNLP 2025**, 20542–20564.

C25. Utkarsh Arora, Anupam Garg, Aryan Gupta, Samyak Jain, Ronit Mehta, Rupin Oberoi, Prachi, Aryaman Raina, Manav Saini, Sachin Sharma, Jaskaran Singh, Sarthak Tyagi, and **Dhruv Kumar**. 2025. Analyzing LLM Usage in an Advanced Computing Class in India. In **ACE '25**, 154–163.

C26. Ritvik Budhiraja, Ishika Joshi, Jagat Sesh Challa, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. "It's not like Jarvis, but it's pretty close!"—Examining ChatGPT's Usage among Undergraduate Students in CS. In **ACE 2024**, 124–133.

C27. Shubhang Shekhar Dvivedi, Vyshnav Vijay, Sai Leela Rahul Pujari, Shoumik Lodh, and **Dhruv Kumar**. 2024. A Comparative Analysis of Large Language Models for Code Documentation Generation. In **AIware 2024**.

C28. Harshal Dev, Raj Gupta, and **Dhruv Kumar**. 2024. From Cash to Cashless: UPI's Impact on Spending Behavior among Indian Users. In **CHI EA 2024**, 168:1–168:10.

C29. Shikhar Sharma, Manas Mhasakar, Apurv Mehra, Utkarsh Venaik, Ujjwal Singhal, **Dhruv Kumar**, and Kashish Mittal. 2024. Comuniqa: Exploring Large Language Models For Improving English Speaking Skills. In **COMPASS 2024**, 256–267.

C30. Ishika Joshi, Ritvik Budhiraja, Harshal Dev, Jahnvi Kadia, Mohammad Osama Ataullah, Sayan Mitra, Harshal D. Akolekar, and **Dhruv Kumar**. 2024. ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate CS Questions. In **SIGCSE 2024**, 625–631.

C31. **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin: Optimizing Geo-Distributed Joins using Aggregation-Based Transformations. In **CCGrid 2023**, 414–425. **[Best Paper Award]**

C32. Vikram Goyal, Naveen Kumar, Sourav S. Bhowmick, Pawan Goyal, Navneet Goyal, and **Dhruv Kumar** (Eds.). 2023. Big Data and Artificial Intelligence—11th International Conference, **BDA 2023**, Proceedings. Lecture Notes in Computer Science 14418. Springer.

C33. Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, Yuanli Wang, and Abhishek Chandra. 2022. HACCS: Heterogeneity-Aware Clustered Client Selection for Accelerated Federated Learning. In **IPDPS 2022**, 985–995.

C34. Rallapalli Srinivas, Matt Drewitz, Joe Magner, Ajit Pratap Singh, **Dhruv Kumar**, and Yashwant Bhaskar Katpatal. 2022. Simulating Landscape Hydrologic Connectivity in a Precise Manner Using Hydro-Conditioning. In **Advances in Computational Modeling and Simulation**, Lecture Notes in Mechanical Engineering. Springer.

C35. **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2021. AggNet: Cost-Aware Aggregation Networks for Geo-Distributed Streaming Analytics. In **SEC 2021**, 297–311.

C36. Sonal Kumari, Poonam Goyal, Ankit Sood, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2017. Exact, Fast and Scalable Parallel DBSCAN for Commodity Platforms. In **ICDCN 2017**, Article 14.

C37. Poonam Goyal, Sonal Kumari, Sumit Sharma, **Dhruv Kumar**, Vivek Kishore, Sundar Balasubramaniam, and Navneet Goyal. 2016. A Fast, Scalable SLINK Algorithm for Commodity Cluster Computing Exploiting Spatial Locality. In **HPCC 2016**, 268–275.

C38. Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, Navneet Goyal, Saiyedul Islam, and Jagat Sesh Challa. 2015. Parallelizing OPTICS for Commodity Clusters. In **ICDCN 2015**, 33:1–33:10.

C39. Poonam Goyal, Sonal Kumari, **Dhruv Kumar**, Sundar Balasubramaniam, and Navneet Goyal. 2014. Parallelizing OPTICS for Multicore Systems. In **ACM COMPUTE 2014**, 17:1–17:6.

Workshop Papers, Posters & Abstracts

W1. Aviral Gupta, Armaan Sethi, and **Dhruv Kumar**. 2026. TabPFN Through The Looking Glass: An Interpretability Study of TabPFN and Its Internal Representations. In **NeurIPS 2026 Workshop (XAI4Science)**.

W2. Shoyam Kumar Mishra and **Dhruv Kumar**. 2026. Persistent Data-Order Effects Precede Directional Alignment in Language Models. In **NeurIPS 2026 Workshop (DynaFront)**.

W3. Vaibhav Devraj, **Dhruv Kumar**, Jagat Sesh Challa, Parth Agarwal, Navya Kommuri, Trizal Garg, Prisha Singhal, Dhruv Shah, and Yash Sinha. 2026. CricBench: A Multilingual Benchmark for Evaluating LLMs in Cricket Analytics. In **EMNLP 2026 Workshop (REALM)**.

W4. Parth Agarwal, Sanchit Bahuguna, and **Dhruv Kumar**. 2026. LLM Graders Are Systematically Harsher Than Industry Interviewers: A Benchmark of 221 Real Technical Interviews. In **EMNLP 2026 Workshop (REALM)**.

W5. Yash Sinha, Parth Shrikant Patil, Murari Mandal, and **Dhruv Kumar**. 2026. Beyond Accuracy: Diagnosing Algebraic Reasoning Failures in LLMs Across Nine Complexity Dimensions. In **EMNLP 2026 Workshop (MathNLP)**.

W6. Vedant Jawandhia, Yash Sinha, Ankan Pal, Murari Mandal, and **Dhruv Kumar**. 2026. Measuring Representation Robustness in Large Language Models for Geometry. In **ICML 2026 Workshop (AI4Math)**.

W7. Manan Gupta, Chinmay Pushkar, Sanchit Kabra, **Dhruv Kumar**, and Jagat Sesh Challa. 2026. MultiVulnBench: A Large-Scale Benchmark for Count Bias in LLM-Based Multi-Vulnerability Detection. In **ICML 2026 Workshop (CTB)**, also accepted at **AgenticUQ**, **CompLearn**, **FAGEN**, and **DL4C**.

W8. Ojaswi Prakash, **Dhruv Kumar**, Murari Mandal, Mohan Kankanhalli, and Yash Sinha. 2026. MafiaPersona: A Multi-Agent Adversarial Benchmark for Evaluating Persona Persistence in Large Language Models. In **ICML 2026 Workshop (NExT-Game)**.

W9. Prapti Patra and **Dhruv Kumar**. 2026. PoolBench: Benchmarking Large Language Models on Continuous Physical Action Selection in Eight-Ball Pool. In **ICML 2026 Workshop (NExT-Game)**.

W10. Preetham Sivalingam, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. TSQueryBench: LLM-as-a-Judge for Time-Series Explanations. In **ICML 2026 Workshop (FMSD)**.

W11. Moulik Gupta, Aaditya Jain, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. ReGeN: Reference-Guided Synthetic Multivariate Time Series Generation for Forecasting. In **ICML 2026 Workshop (FMSD)**.

W12. Atharva Gupta, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. Where Computation Lives Inside TabPFN: Causal Localisation of Attention Head Function. In **ICML 2026 Workshop (FMSD)**.

W13. Jay Agarwal, Siddharth Khare, and **Dhruv Kumar**. 2026. What Do Lorentz-Equivariant Jet Taggers Learn? In **ICML 2026 Workshop (AI4Physics)**.

W14. Shivansh Bibra, **Dhruv Kumar**, Murari Mandal, and Yash Sinha. 2026. Joint Evaluation of Compliance, Planning, and Consistency under Paraphrase: A Relational-Complexity View of Frontier LLMs. In **ICML 2026 Workshop (CTB)**.

W15. Jalluri Mahesh Kumar, Junjunoori Sri Chakri, Yash Kothari, Murari Mandal, Yash Sinha, and **Dhruv Kumar**. 2026. Evaluating LLM Reasoning on Operating System Algorithms via Step-Level Verification. In **ICML 2026 Workshop (CTB)**.

W16. Kanav Kapoor, Jagat Sesh Challa, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2026. GraphStateEval: A Step-by-Step Evaluation Framework for Graph Algorithm Execution in Large Language Models via Intermediate State Tracing. In **ICML 2026 Workshop (CTB)**.

W17. Shubh Chapra, **Dhruv Kumar**, Murari Mandal, and Yash Sinha. 2026. The Complexity Ceiling Benchmark: A Multi-Domain Evaluation of Sequential Reasoning Under Depth Scaling. In **ICML 2026 Workshop (CTB)**.

W18. Aryan Khurana, Aravind Ramana RN, and **Dhruv Kumar**. 2026. Authority, Truth, and Citation Bias: A Large-Scale Multi-Domain Benchmark for Studying Epistemic Susceptibility in Large Language Models. In **ICML 2026 Workshop (EIML)**, also accepted at **AI4GOOD** and **NeurIPS 2026 Workshop (AI4GOOD)**.

W19. Agniva Maiti, Aditya Tiwari, Dwarikanath Mahapatra, Sudipta Roy, Yash Sinha, **Dhruv Kumar**, and Murari Mandal. 2026. Bridging the Visual Gap: Resource-Efficient VLM Adaptation for Meitei Mayek using Synthetic Multi-Modal Mixtures. In **ICML 2026 Workshop (GlobalSouthML)**.

W20. Agniva Maiti, Aarsh Muley, Aditya Tiwari, Dwarikanath Mahapatra, Sudipta Roy, Yash Sinha, **Dhruv Kumar**, and Murari Mandal. 2026. Evaluating Indic Literary Creativity In LLMs through the Lens of the Shastras. In **ICML 2026 Workshop (GlobalSouthML)**.

W21. Aviral Dawar, Roshan Karanth, Vikram Goyal, and **Dhruv Kumar**. 2026. IndicDB: Benchmarking Multilingual Text-to-SQL Capabilities in Indian Languages. In **ICML 2026 Workshop (Culture x AI)**, also accepted at **GlobalSouthML** and **ACL 2026 Workshop (MeLLM)**.

W22. Manya Pandey, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2026. GITCO: Gated Inference-Time Context Optimization in TSFMs. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD**.

W23. Aaryan Nagpal, Debdeep Sanyal, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. Mix, Don't Pick: Why Synthetic Corpus Composition Matters for Time Series Foundation Model Pretraining. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD** and **Forecast**.

W24. Karthik Sridhar, Atharva Gupta, Nishant Pradhan, Murari Mandal, **Dhruv Kumar**, and Saurabh Deshpande. 2026. Semantics or Structure? Auditing Text Sensitivity in Multimodal Time-Series Forecasting. In **ICML 2026 Workshop (GlobalSouthML)**, also accepted at **FMSD**.

W25. Manan Gupta and **Dhruv Kumar**. 2026. Diagnosing LLM Judge Reliability: Conformal Prediction Sets and Transitivity Violations. In **ICML 2026 Workshop (SCALE)**, also accepted at **CTB**, **SPIGM**, **FoGen**, and **FAGEN**.

W26. Manan Gupta and **Dhruv Kumar**. 2026. Is the Trust Region Necessary? Robust Entropy Adaptation Suffices for Continuous Control. In **ICML 2026 Workshop (SCALE)**.

W27. Manan Gupta and **Dhruv Kumar**. 2026. Latent Phase-Shift Rollback: Inference-Time Error Correction via Residual Stream Monitoring and KV-Cache Steering. In **ICML 2026 Workshop (SCALE)**, also accepted at **AgenticUQ**, **CTB**, **FoGen**, and **FAGEN**.

W28. Manan Gupta and **Dhruv Kumar**. 2026. Reasoning Phases Are Continuous, Not Discrete: Evidence from Switching Linear Dynamical Systems Applied to Chain-of-Thought Residual Streams. In **ICML 2026 Workshop (SCALE)**, also accepted at **CompLearn**, **FoGen**, and **Mech Interp**.

W29. Manan Gupta, Inderjeet Jayakumar Nair, Lu Wang, and **Dhruv Kumar**. 2026. Context Over Content: Exposing Evaluation Faking in Automated Judges. In **ICML 2026 Workshop (AgenticUQ)**, also accepted at **SPIGM**, **FoGen**, **FAGEN**, and **AI4GOOD**.

W30. Manan Gupta and **Dhruv Kumar**. 2026. DPMI: A Principled Index for Neural Polysemanticity via Dirichlet Process Mixture Modeling. In **ICML 2026 Workshop (SCALE)**, also accepted at **CompLearn** and **FoGen**.

W31. Aaryaman Kalani, **Dhruv Kumar**, Mohan Kankanhalli, Murari Mandal, and Yash Sinha. 2026. RASE: Retain-Agnostic Machine Unlearning via Activation Subspace Projection. In **CVPR 2026 Workshop (MUV)**.

W32. Shubham Mishra, Shiv Tiwari, Samyek Jain, Gorang Mehrishi, **Dhruv Kumar**, Pratik Narang, and Harsh Sharma. 2026. From Facts to Conclusions: Integrating Deductive Reasoning in Retrieval-Augmented LLMs. In **ICLR 2026 Workshop on LLM Reasoning**. **arXiv (2025)**.

W33. Debdeep Sanyal, Aaryan Nagpal, **Dhruv Kumar**, Murari Mandal, and Saurabh Deshpande. 2025. time2time: Causal Intervention in Hidden States to Simulate Rare Events in Time Series Foundation Models. In **NeurIPS 2025 Workshop on Time Series Foundation Models (BERT4S)**.

W34. Sankalp Tattwadarshi Swain, Anshika Krishnatray, **Dhruv Kumar**, and Jagat Sesh Challa. 2025. Talking with Oompa Loompas: A Novel Framework for Evaluating Linguistic Acquisition of LLM Agents. In **NeurIPS 2025 Workshop (WiML)**, also accepted at **LLM Evaluation**.

W35. Utkarsh Tiwari, Aryan Seth, Adi Mukherjee, Kaavya Mer, Kavish, and **Dhruv Kumar**. 2025. DebateBench: A Challenging Long Context Reasoning Benchmark For Large Language Models. In **NAACL SRW 2025**.

W36. Shreya Bhatia, Tarushi Gandhi, **Dhruv Kumar**, and Pankaj Jalote. 2024. Unit Test Generation using Generative AI: A Comparative Performance Analysis of Autogeneration Tools. In **LLM4Code@ICSE 2024**, 54–61.

W37. **Dhruv Kumar**, Sohaib Ahmad, Abhishek Chandra, and Ramesh K. Sitaraman. 2023. AggFirstJoin (Workshop Version). In **CCGridW 2023**, 308–310.

W38. **Dhruv Kumar**, Joel Wolfrath, Abhishek Chandra, and Ramesh K. Sitaraman. 2022. Towards WAN-Aware Join Sampling over Geo-Distributed Data. In **EdgeSys@EuroSys 2022**, 13–18.

W39. Yuanli Wang, Joel Wolfrath, Nikhil Sreekumar, **Dhruv Kumar**, and Abhishek Chandra. 2021. Accelerated Training via

Device Similarity in Federated Learning. In **EdgeSys@EuroSys 2021**, 31–36.

W40. Yuanli Wang, **Dhruv Kumar**, and Abhishek Chandra. 2020. Exploiting Data Heterogeneity for Performance and Reliability in Federated Learning (Poster). In **SEC 2020**, 164–166.

W41. **Dhruv Kumar**, Aravind Alagiri Ramkumar, Rohit Sindhu, and Abhishek Chandra. 2019. DeCaf: Iterative Collaborative Processing over the Edge. In **HotEdge 2019**.

W42. **Dhruv Kumar**, Jian Li, Abhishek Chandra, and Ramesh K. Sitaraman. 2019. A TTL-Based Approach for Data Aggregation in Geo-Distributed Streaming Analytics (Abstract). In **SIGMETRICS 2019**, 15–16.

Preprints (arXiv / SSRN)

P1. Rakshita Goel, S. Pranav Kumar, Anmol Agrawal, Divyan Poddar, Pratik Narang, and **Dhruv Kumar**. 2026. Domain-Partitioned Hybrid RAG for Legal Reasoning: Toward Modular and Explainable Legal AI for India. **arXiv (2026)**.

P2. Gaurish Trivedi, Alakh Sharma, Kartikey Singh Bhandari, Yash Sinha, Pratik Narang, **Dhruv Kumar**, and Jagat Sesh Challa. 2026. Trust Regions Sell, But Who's Buying? Overlap Geometry as an Alternative Trust Region for Policy Optimization. **arXiv (2026)**.

P3. Devanshu Sahoo, Manish Prasad, Vasudev Majhi, Arjun Neekhra, Yash Sinha, Murari Mandal, Vinay Chamola, and **Dhruv Kumar**. 2026. The Compliance Paradox: Semantic-Instruction Decoupling in Automated Academic Code Evaluation. **arXiv (2026)**.

P4. Aditya Thole, Anmol Agrawal, Arnav Ramamoorthy, and **Dhruv Kumar**. 2026. PhysicsSolutionAgent: Towards Multimodal Explanations for Numerical Physics Problem Solving. **arXiv (2026)**.

P5. Kartikey Singh Bhandari, Manav Ganesh, Yashwant Viswanathan, Archit Agrawal, **Dhruv Kumar**, and Pratik Narang. 2026. Actionable Advice from Reviews via Mixture of LoRA Experts: A Two-LLM Pipeline for Issue Extraction and Business Recommendations. **arXiv (2026)**.

P6. Kartikey Singh Bhandari, Tanish Jain, Archit Agrawal, **Dhruv Kumar**, Praveen Kumar, and Pratik Narang. 2026. A Multi-Agent System for Generating Actionable Business Advice. **arXiv (2026)**.

P7. Devesh Saraogi, Rohit Singhee, and **Dhruv Kumar**. 2026. Evaluating Novelty in AI-Generated Research Plans Using Multi-Workflow LLM Pipelines. **arXiv (2026)**.

P8. Varun Bharti, Shashwat Jha, **Dhruv Kumar**, and Pankaj Jalote. 2025. Automated Type Annotation in Python Using Large Language Models. **arXiv:2508.00422 (2025)**.

P9. Gaurish Trivedi, Alakh Sharma, Kartikey Singh Bhandari, **Dhruv Kumar**, Pratik Narang, and Jagat Sesh Challa. 2025. HAEPO: History-Aggregated Exploratory Policy Optimization. **arXiv:2508.18884 (2025)**.

P10. Manan Gupta and **Dhruv Kumar**. 2025. Disentangling Polysemantic Neurons with a Null-Calibrated Polysemanticity Index and Causal Patch Interventions. **arXiv:2508.16950 (2025)**.

P11. Unnati Saraswat, Tarun Rao, Namah Gupta, Shweta Swami, Shikhar Sharma, Prateek Narang, and **Dhruv Kumar**. 2025. Toward Intelligent Scene Augmentation for Context-Aware Object Placement and Sponsor-Logo Integration. **arXiv (2025)**.

P12. Arnav Gupta, Gurekas Singh Sahney, Hardik Rathi, Abhishek Chandwani, Ishaan Gupta, Pratik Narang, and **Dhruv Kumar**. 2025. Understanding Virality: A Rubric-Based Vision-Language Model Framework for Short-Form Edutainment Evaluation. **arXiv (2025)**.

P13. Akshaj Prashanth Rao, Advait Singh, Saumya Kumaar Saksena, and **Dhruv Kumar**. 2025. Efficient Jailbreak Mitigation Using Semantic Linear Classification in a Multi-Staged Pipeline. **arXiv (2025)**.

P14. Aryan Chaudhary, Sanchit Goyal, Pratik Narang, and **Dhruv Kumar**. 2025. Investigating Spatial Attention Bias in Vision-Language Models. **arXiv (2025)**.

P15. Daksh Jain, Aarya Jain, Ashutosh Desai, Avyakt Verma, Ishan Bhanuka, Pratik Narang, and **Dhruv Kumar**. 2025. Large Language Models as Pokémon Battle Agents: Strategic Play and Content Generation. **arXiv (2025)**.

P16. Chayan Jain, Rishant Sharma, Archit Garg, Ishan Bhanuka, Pratik Narang, and **Dhruv Kumar**. 2025. Lights, Camera, Consistency: A Multistage Pipeline for Character-Stable AI Video Stories. **arXiv (2025)**.

P17. Charan Prakash Rathore, Saumi Ray, and **Dhruv Kumar**. 2025. Evaluating LLMs for Zeolite Synthesis Event Extraction (ZSEE): A Systematic Analysis of Prompting Strategies. **arXiv (2025)**.

P18. Arnav Ramamoorthy, Shrey Dhorajiya, Ojas Pungalia, Rashi Upadhyay, Abhishek Mishra, Abhiram H., Tejasvi Alladi, Sujan Yenuganti, and **Dhruv Kumar**. 2025. neuralFOMO: Can LLMs Handle Being Second Best? Measuring Envy-Like Preferences in Multi-Agent Settings. **arXiv (2025)**.

P19. Devanshu Sahoo, Vasudev Majhi, Arjun Neekhra, Yash Sinha, Murari Mandal, and **Dhruv Kumar**. 2025. How to Trick Your AI TA: A Systematic Study of Academic Jailbreaking in LLM Code Evaluation. **arXiv (2025)**.

P20. Madhav Krishna, Dhruv Gupta, Kotra Keshav, Hritav Singh, Akash Bonagiri, Pratik Narang, and **Dhruv Kumar**. 2025. MarketMaven: An Agent-Based AI for Automated Startup Analysis and Market Intelligence. **SSRN (2025)**.

P21. Sasank Tumpati, Devansh Agarwal, and **Dhruv Kumar**. 2025. Magistrate: A Repository-Context Aware Multi-Agent Framework for Automated Code Review. **SSRN (2025)**.

P22. Debdeep Sanyal, Manya Pandey, **Dhruv Kumar**, Saurabh Deshpande, and Murari Mandal. 2025. Confidence is Not Competence: Evaluating Calibration in LLMs. **arXiv (2025)**.

P23. Henansh Tanwar, Kunal Shrivastva, Rahul Singh, and **Dhruv Kumar**. 2024. OpineBot: Class Feedback Reimagined Using a Conversational LLM. **arXiv:2401.15589 (2024)**.

P24. Aryaman Raina, Prateek Mishra, Harshit Goyal, and **Dhruv Kumar**. 2024. AI as a Medical Ally: Evaluating ChatGPT's Usage and Impact in Indian Healthcare. **arXiv:2401.15605 (2024)**.

P25. Vibhor Agarwal, Nakul Thureja, Madhav Krishan Garg, Sahiti Dharmavaram, Meghna, and **Dhruv Kumar**. 2024. "Which LLM should I use?": Evaluating LLMs for Tasks Performed by Undergraduate CS Students in India. **arXiv:2402.01687 (2024)**.

P26. Anishka, Diksha Sethi, Nipun Gupta, Shikhar Sharma, Srishti Jain, Ujjwal Singhal, and **Dhruv Kumar**. 2024. TAMIGO: Empowering Teaching Assistants using LLM-Assisted Viva and Code Assessment. **arXiv:2407.16805 (2024)**.

P27. Shreya Bhatia, Tarushi Gandhi, **Dhruv Kumar**, and Pankaj Jalote. 2024. System Test Case Design from Requirements Specifications: Insights and Challenges of Using ChatGPT. **arXiv:2412.03693 (2024)**.

P28. Ishika Joshi, Ritvik Budhiraja, Pranav Deepak Tanna, Lovenya Jain, Mihika Deshpande, Arjun Srivastava, Srinivas Rallapalli, Harshal D. Akolekar, Jagat Sesh Challa, and **Dhruv Kumar**. 2023. "With Great Power Comes Great Responsibility!": Student and Instructor Perspectives on the Influence of LLMs on Undergraduate Engineering Education. **arXiv:2309.10694 (2023)**.

P29. Atharva Mehta, Nipun Gupta, Aarav Balachandran, **Dhruv Kumar**, Pankaj Jalote, et al. 2023. Can ChatGPT Play the Role of a Teaching Assistant in an Introductory Programming Course? **arXiv (2023)**.

Sponsored Research Projects

1. SPANDAN — AI/ML for Strengthening Police Investigation Processes

Funding Agency	Gujarat Police (Office of Superintendent of Police, Ahmedabad Rural)
Budget	INR 34,80,000
Duration	11 months (03/2026–01/2027)
Status	Ongoing
Role	PI

2. Revolutionizing Indian Higher Education Institutes with Generative AI

Funding Agency	BITS Pilani
Budget	INR 20,00,000
Duration	24 months (11/2024–10/2026)
Status	Ongoing
Role	PI

Summary of Results & Publications:

- Pathak, A.; **Kumar, D.**; et al. 2025. Rubric Is All You Need: LLM-based code evaluation. *ACM ICER 2025*.
- Arora, U.; **Kumar, D.**; et al. 2025. Analyzing LLM usage in an advanced computing class in India. *ACE '25*, 154–163.

3. AI for River–Climate Modeling

Funding Agency	The Nature Conservancy (NGO)
Budget	INR 25,00,000
Duration	11 months (08/2024–06/2025)
Status	Completed
Role	Co-PI

Summary of Results & Publications:

- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the role of land use–land cover for adaptive river basin management. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Targeting degraded hotspots in riparian corridors for rehabilitation. *Journal of Environmental Management* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Quantifying the synergistic influence of LULC and climate on hydro-climatological dynamics. *Journal of Hydrology* (under submission).
- Rallapalli, S.; **Kumar, D.**; et al. 2025. Disaggregating the influence of climate change and LULC on streamflow. *Journal of Environmental Management* (under submission).

4. Generative AI in Education Grant

Funding Agency	WISE–IIE Higher Education Consortium
Budget	INR 30,00,000
Duration	10 months (08/2024–05/2025)
Status	Completed
Role	Co-PI

Summary of Results & Publications:

- Akolekar, H.; **Kumar, D.**; et al. 2025. Generative AI tools in mechanical engineering education. *Scientific Reports* 15, 9214.
- Gupta, R.; **Kumar, D.**; et al. 2025. Sakshm AI: Socratic tutoring for coding education in India. *ACM JCSS 2026*.
- Goyal, H.; **Kumar, D.**; et al. 2025. Impact of LLMs on K-12 education in rural India. *Scientific Reports 2025*.
- Sanyal, D.; **Kumar, D.**; et al. 2025. Investigating pedagogical teacher and student LLM agents. *EMNLP 2025*.

5. Google Cloud Education Grant

Funding Agency	Google Cloud
Budget	INR 12,50,000
Duration	12 months (01/2024–12/2024)
Status	Completed
Role	PI